

```

35
36 # Pack up the parameters and initial
conditions:
37 p = [m1,m2,k1,k2,L1,L2,b1,b2]
38 w0 = [x1,y1,x2,y2]
39
40 # Call the ODE solver.
41 wsol = odeint(two_springs,ve
ctorfield,w0,t,args=(p,),atol=abs
err,rtol=relerr)
42
43 # Print the solution.
44 for t1,w1 in zip(t,wsol):
45     print t1,w1[0],w1[1],w1[2],w1[3]

```

```

4 from pylab import *
5 from matplotlib.font_manager import
FontProperties
6 import numpy as np
7
8 data = np.loadtxt('two_springs.txt')
9 t = data[:,0]
10 x1 = data[:,1]
11 y1 = data[:,2]
12 x2 = data[:,3]
13 y2 = data[:,4]
14
15 figure(1,figsize=(6,4))
16
17 xlabel('t')
18 grid(True)
19 hold(True)
20 lw = 1
21
22 plot(t,x1,'b',linewidth=lw)
23 plot(t,x2,'g',linewidth=lw)
24
25 legend(['$x_{1S}$','$x_{2S}$'],prop=Font
Properties(size=16))
26 title('Mass Displacements for the
Coupled Spring-Mass
System')
27 savefig('two_springs.png',dpi=72)

```

The `scipy.integrate.odeint` function integrates a system of ordinary differential equations. It takes the following parameters:

func: callable(y, t0, ...) - It computes the derivative of y at t0.

y0: array - This is the initial condition only (can be a vector).

t: array - It is a sequence of time points for which to solve for y. The initial value point should be the first element of this sequence.

args: tuple - Indicates extra arguments to pass to function. In our example, we have added `atol` and `rtol` as extra arguments to deal with absolute and relative errors.

The `zip` function takes one or more sequences as arguments, and returns a series of tuples that pair up parallel items taken from those sequences.

Copy the solution generated from this script to a text file using the `cat` command. Name this text file as `two_springs.txt`.

The following script uses Matplotlib to plot the solution generated by `two_springs_solver.py`:

```

1 #!/usr/bin/python
2 # Defining a matrix and matrix
multiplication
3

```

By: Adit Gupta

Adit Gupta works as a technology evangelist at Sourcebits Technologies, Bangalore. He is passionate about programming, astrophysics and computer science. You can find him on Twitter at <http://twitter.com/aditgupta>

STUDY GUIDE

RHCE SERIES

LINUX ESSENTIALS, SERVICES & SECURITY

LEARN BY EXAMPLE



VIPIN GUPTA

Price: US \$40, India Rs. 650/-

This study guide starts from basics & takes you to advance levels of linux. everything has been explained by giving examples. Proper lab setup along with diagrams has been given so that you can construct the labs yourself & do the practice. It covers the course contents of RHCE. It is for those who want to start fresh as well as who have done RHCT/RHCE

It is available in ebook as well as paperback format. paperback edit on is available only through direct order. For more details visit our website or email



About Author: Vipin Gupta is BE, RHCE, CEH, ASM, CCNA, MCSE, MCSA. He has got more than 17 years of experience in IT related fields.. During his long career, he has been associated with various engineering colleges, polytechnics, private institutes & corporate sector.

Courses/Workshops

- Linux Administration
- Linux Security Specialist
- LMS using Moodle
- 1-Day/3-Day/5-Day Workshops

we focus more on
knowledge than on
certifications

we also provide
virtual classroom solutions
for colleges/schools

U-NET SOLUTIONS

Opp. Khalsa School For Boys, MOGA (Pb)
Ph. 93563-10379
vipin2411@gmail.com

www.linux-expert.in